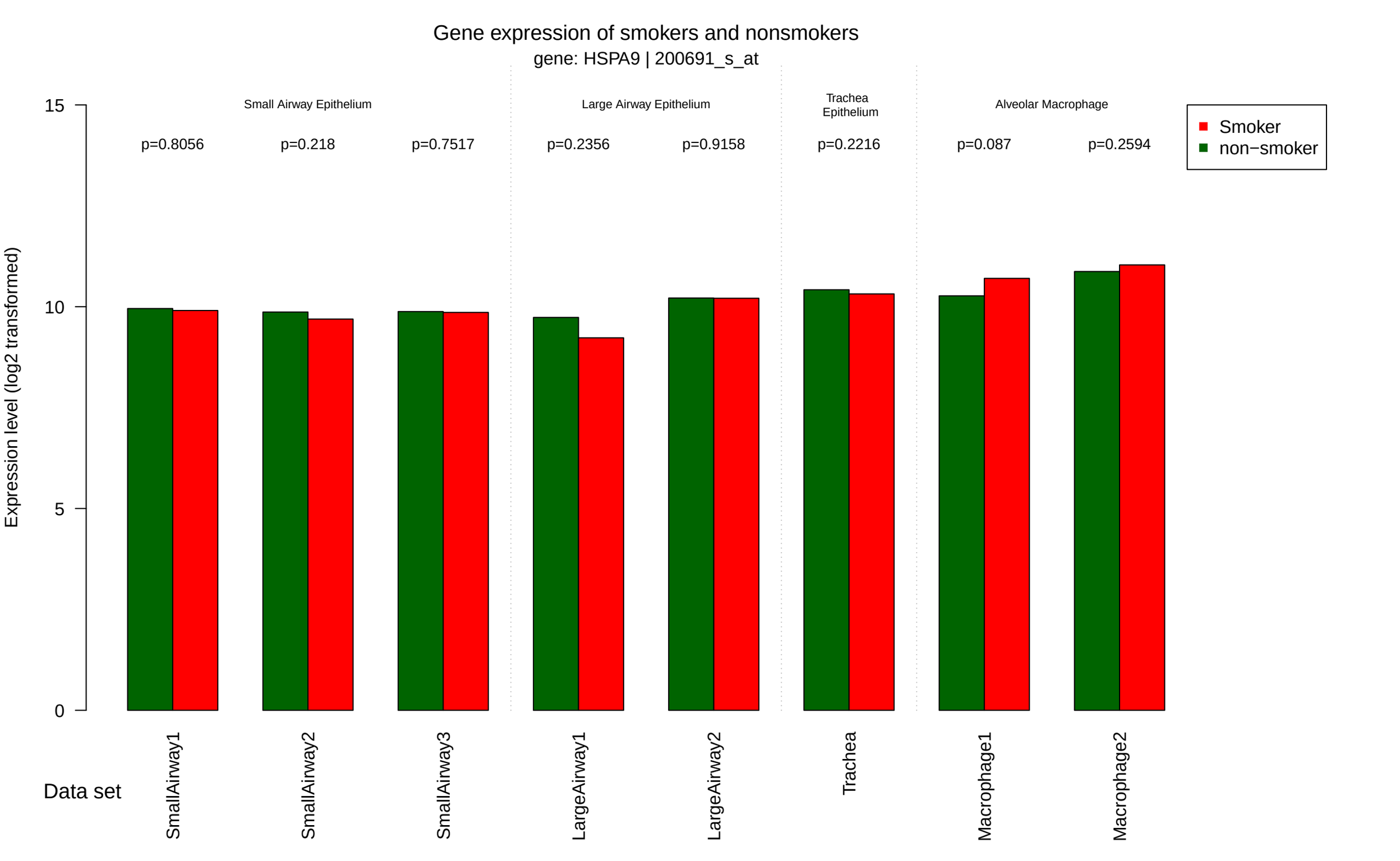
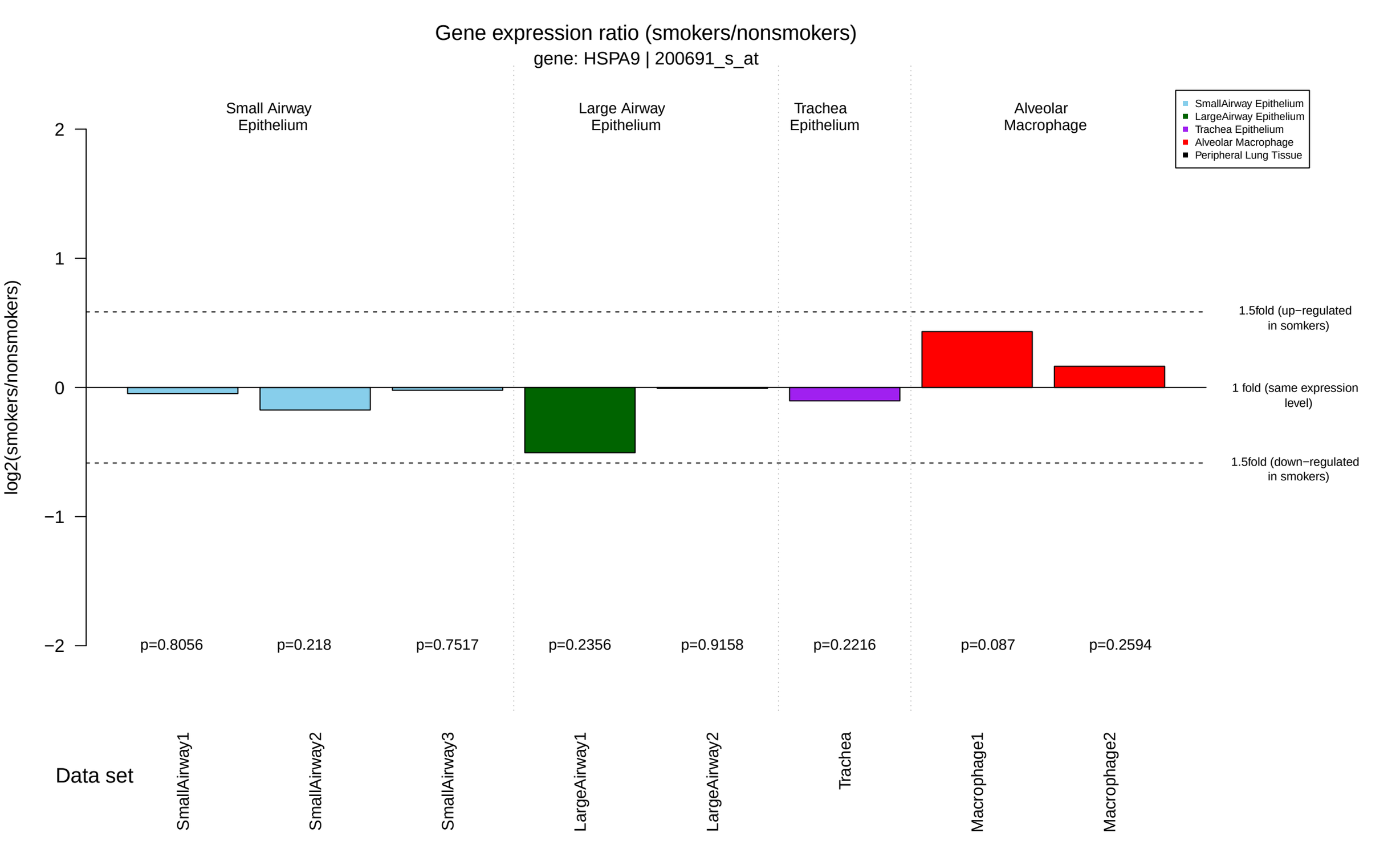
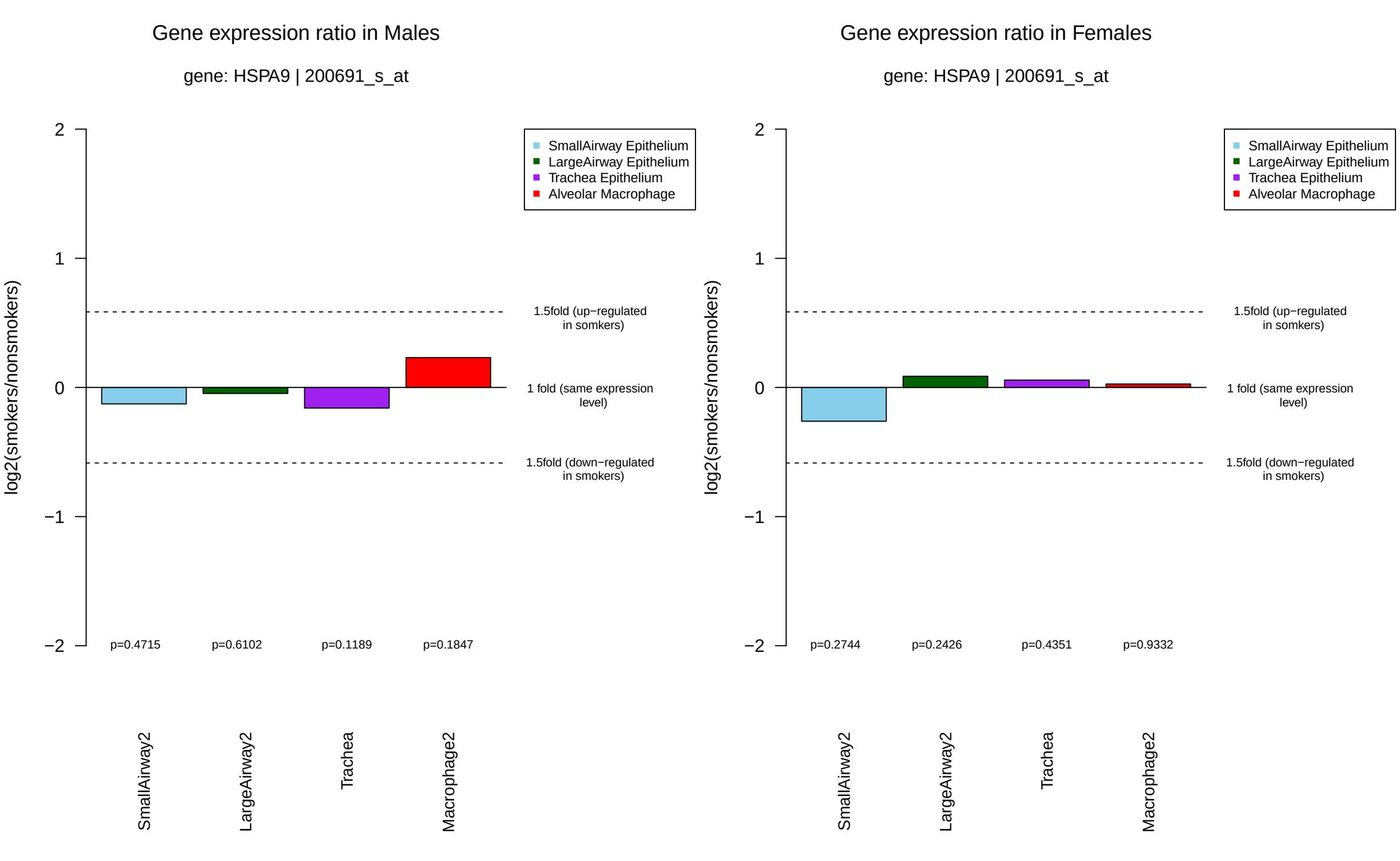
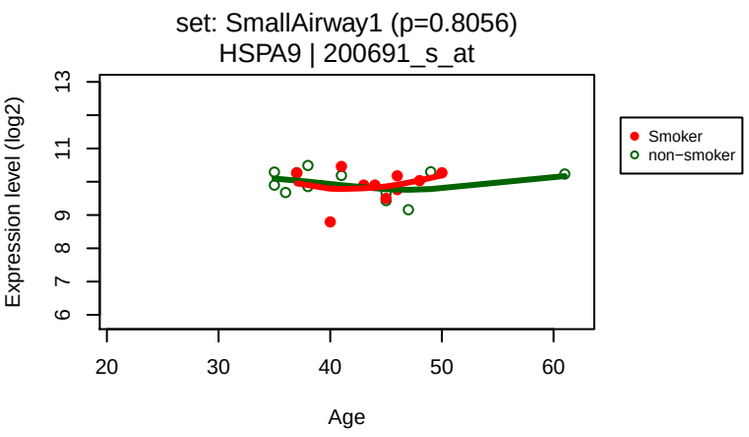
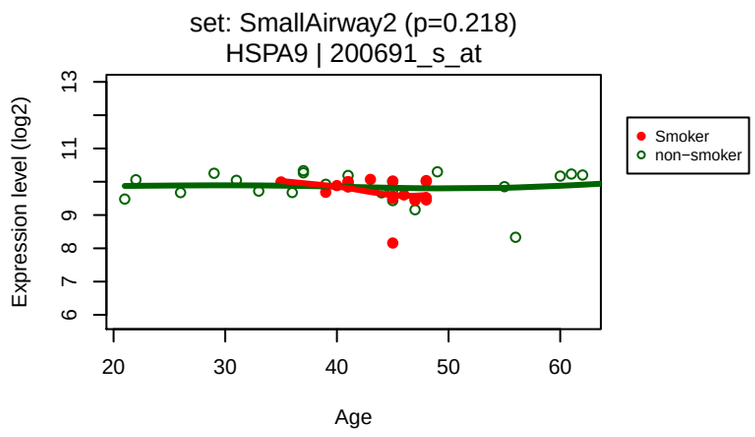
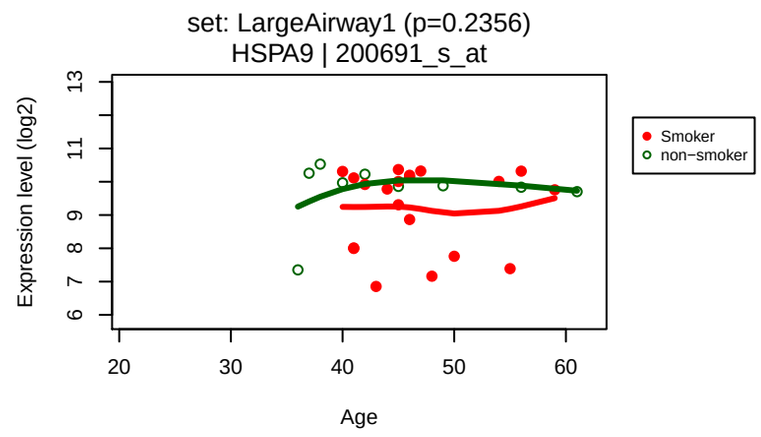

Gene expression of smokers and nonsmokers
gene: HSPA9 | 200691_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8056
p=0.218
p=0.7517
p=0.2356
p=0.9158
p=0.2216
p=0.087
p=0.2594
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

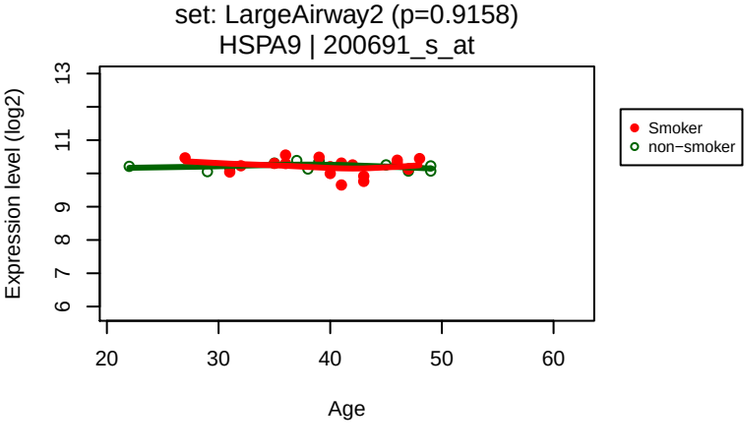

Gene expression ratio (smokers/nonsmokers)
gene: HSPA9 | 200691_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8056
p=0.218
p=0.7517
p=0.2356
p=0.9158
p=0.2216
p=0.087
p=0.2594
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

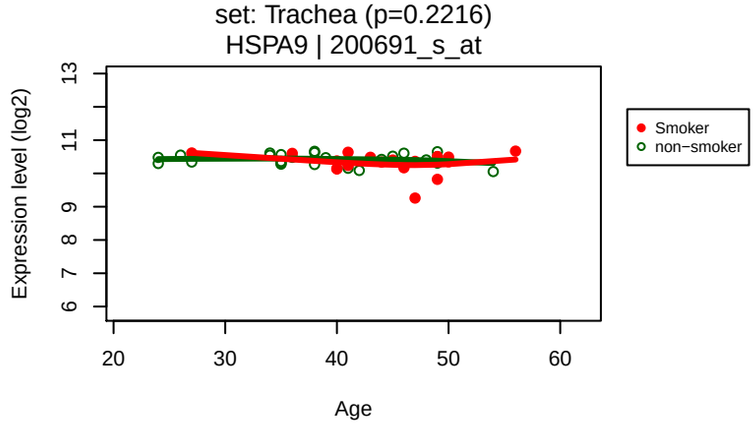

Gene expression ratio in Males
gene: HSPA9 | 200691_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4715
p=0.6102
p=0.1189
p=0.1847
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA9 | 200691_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2744
p=0.2426
p=0.4351
p=0.9332
SmallAirway2
LargeAirway2
Trachea
Macrophage2

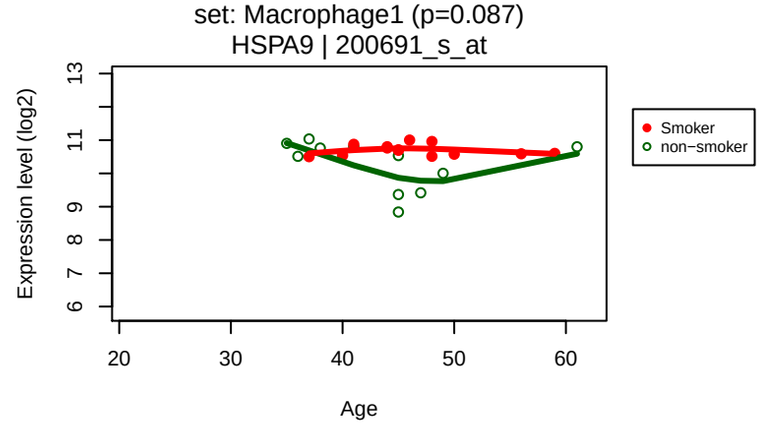

set: SmallAirway1 (p=0.8056)
HSPA9 | 200691_s_at
Expression level (log2)
Age
Smoker
non−smoker

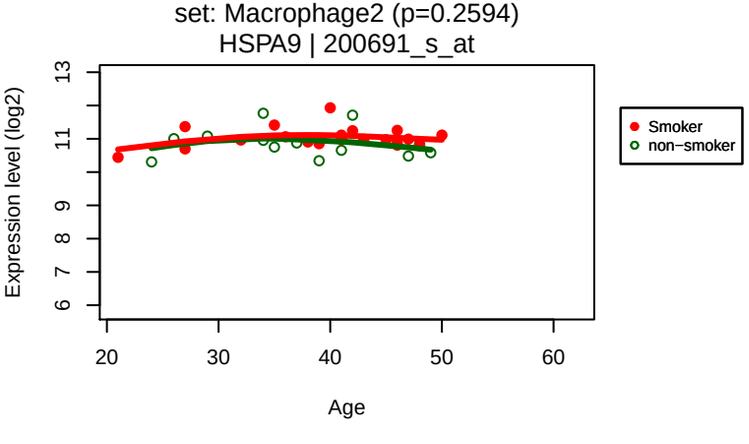

set: SmallAirway2 (p=0.218)
HSPA9 | 200691_s_at
Expression level (log2)
Age
Smoker
non−smoker

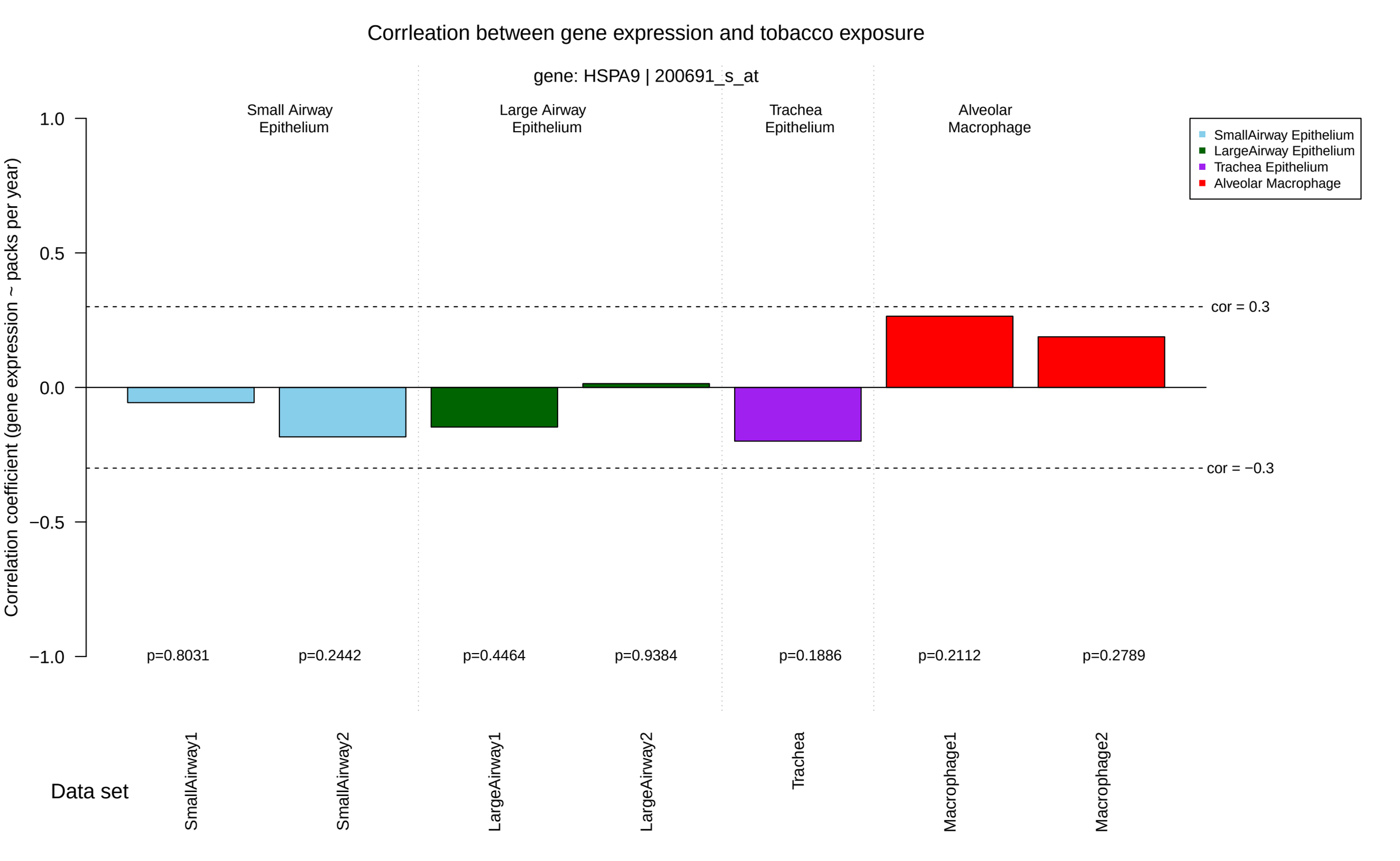

set: LargeAirway1 (p=0.2356)
HSPA9 | 200691_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9158)
HSPA9 | 200691_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2216)
HSPA9 | 200691_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.087)
HSPA9 | 200691_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2594)
HSPA9 | 200691_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA9 | 200691_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8031
p=0.2442
p=0.4464
p=0.9384
p=0.1886
p=0.2112
p=0.2789
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2